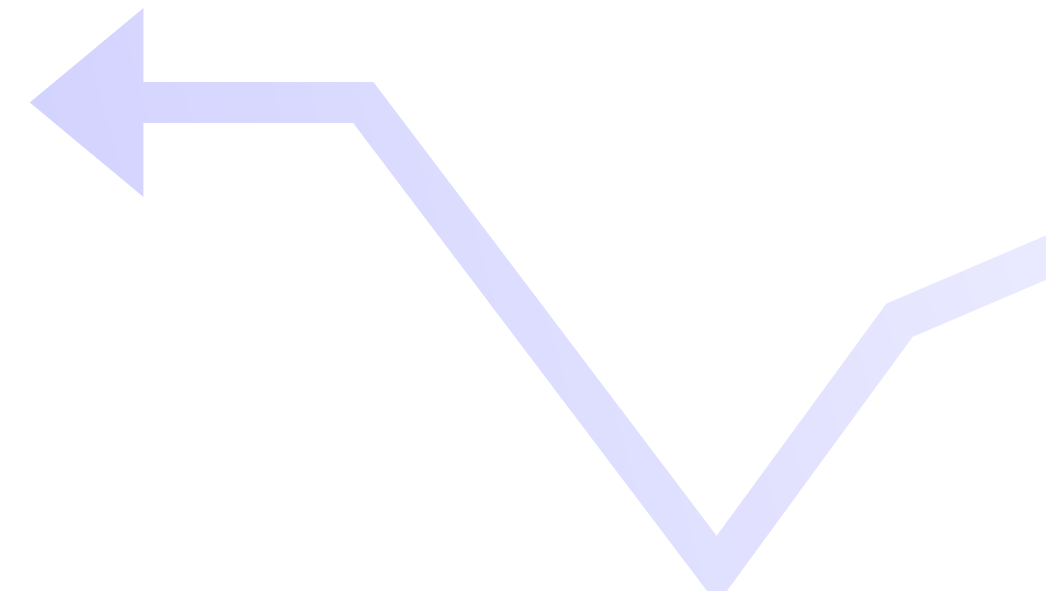
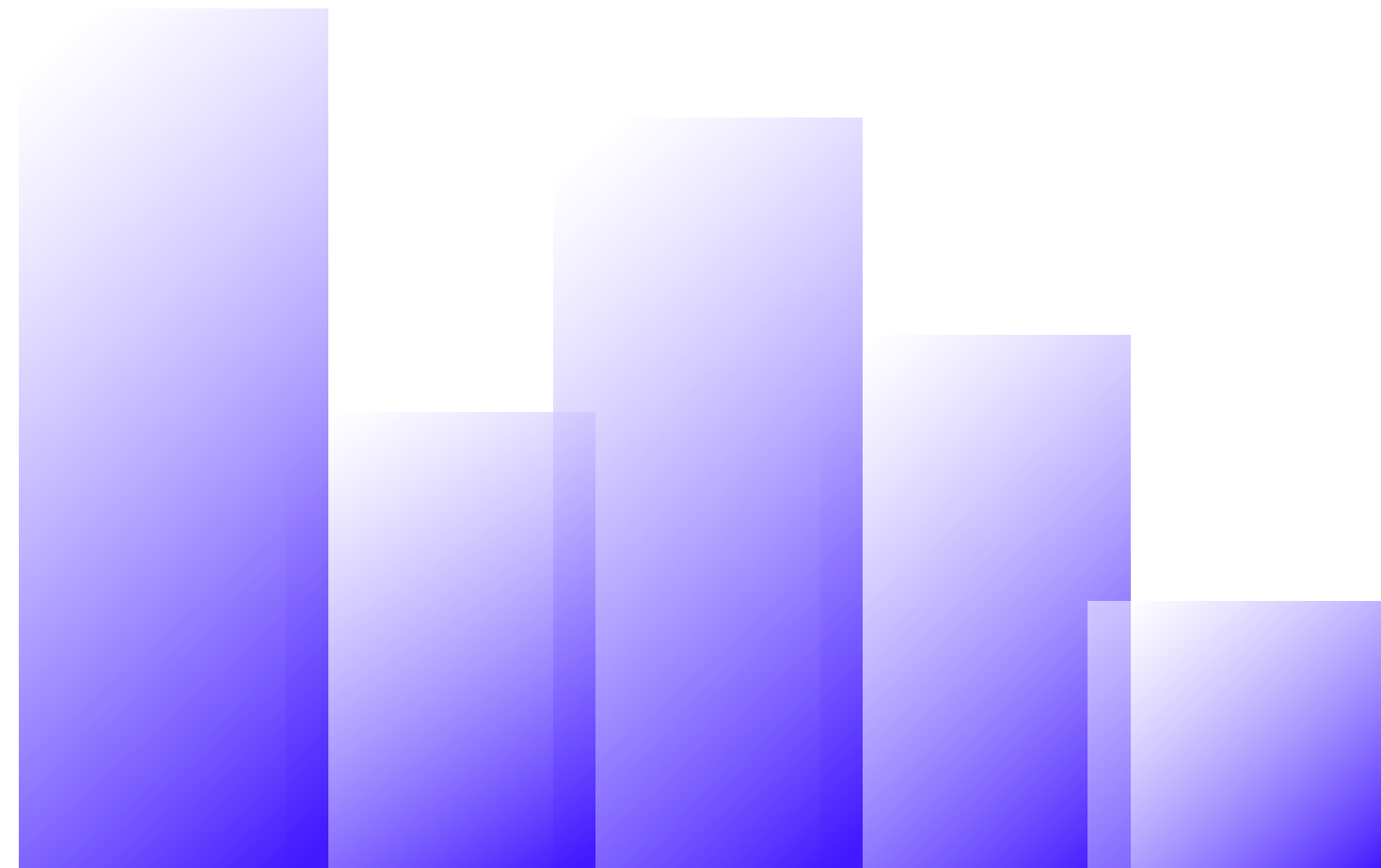




MarketPULSE

Real Time Stock Insights, Trends & Alerts



MarketPULSE

Real Time Stock Insights, Trends & Alerts

By

Purangana Mukhopadhyay (12200122074)

Ankit Lahiri (12200122078)

Amit Samui (12200122095)

Md. Tousif Alam (12200122060)

Under the guidance of

Dr. Tapalina Banerjee Bhattacharjee

Assistant Professor, Department of CSE

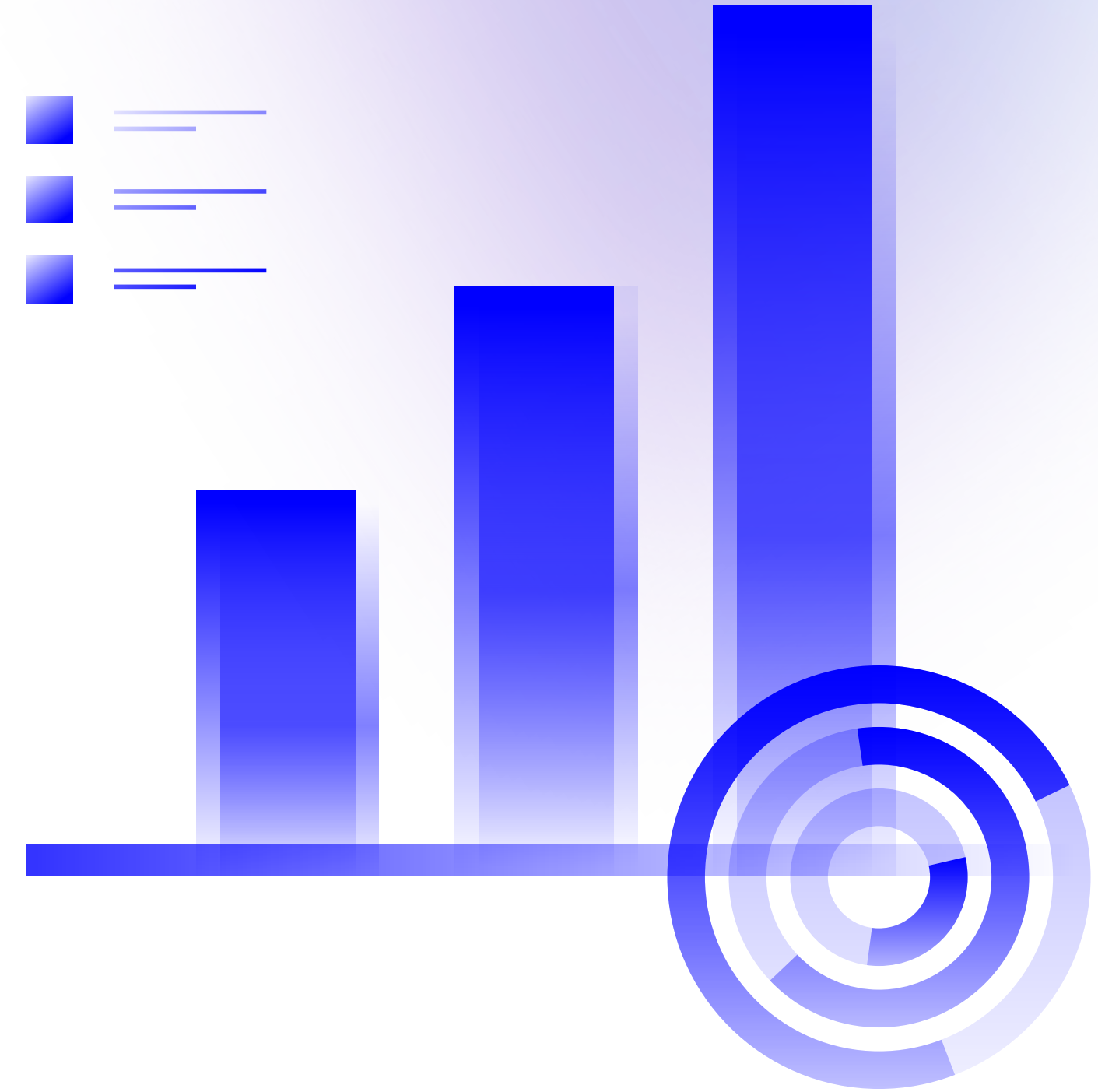
**Department of Computer Science and Engineering
St. Thomas' College of Engineering and Technology**

Affiliated to

Maulana Abul Kalam Azad University of Technology, West Bengal

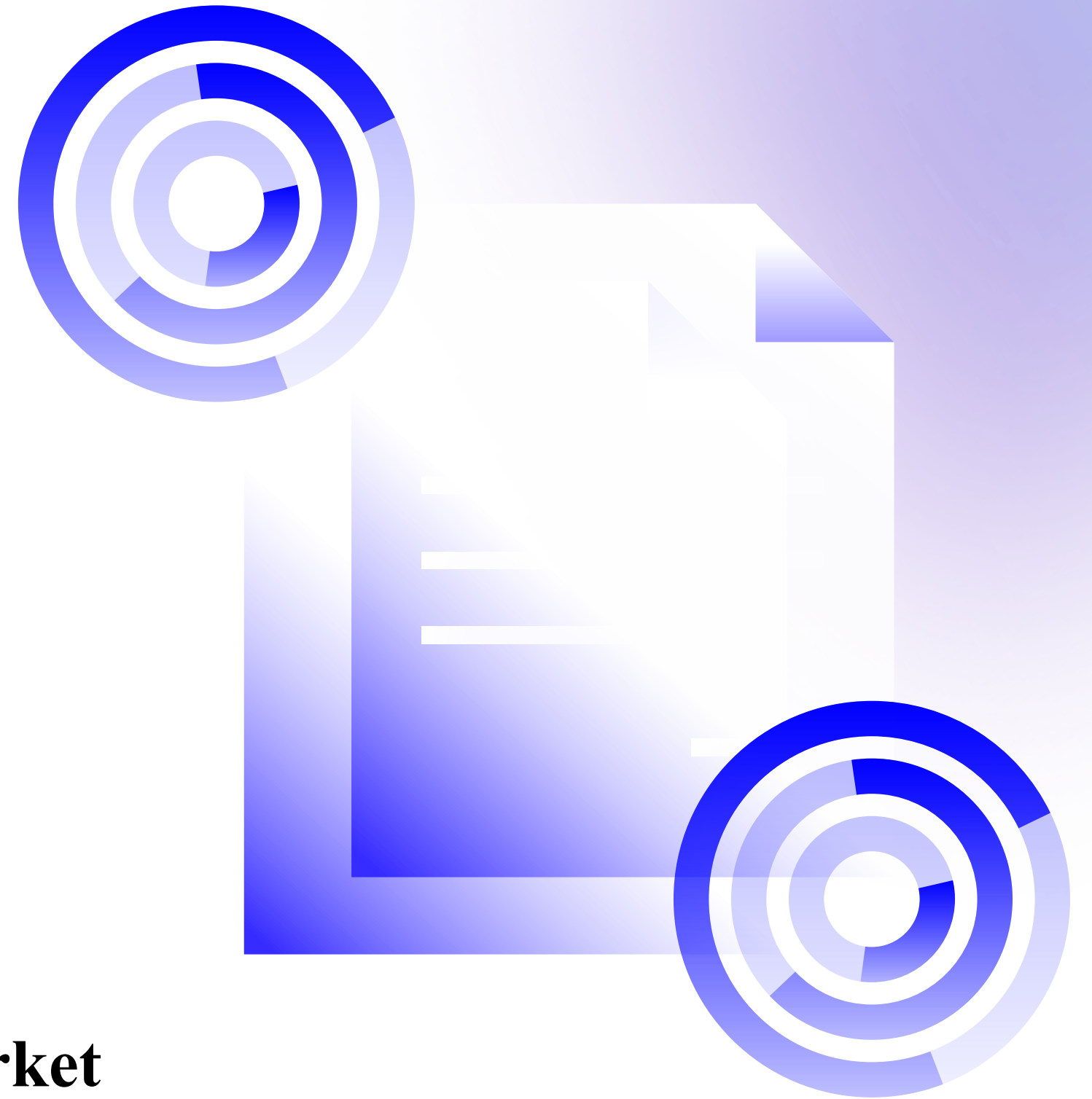
Outline:

- Objectives
- Workflow
- Literature Survey
- Model Selection
- Technical Requirements
- Methodology & Implementation
- Results
- Performance
- Challenges
- Strengths (USPs)
- References

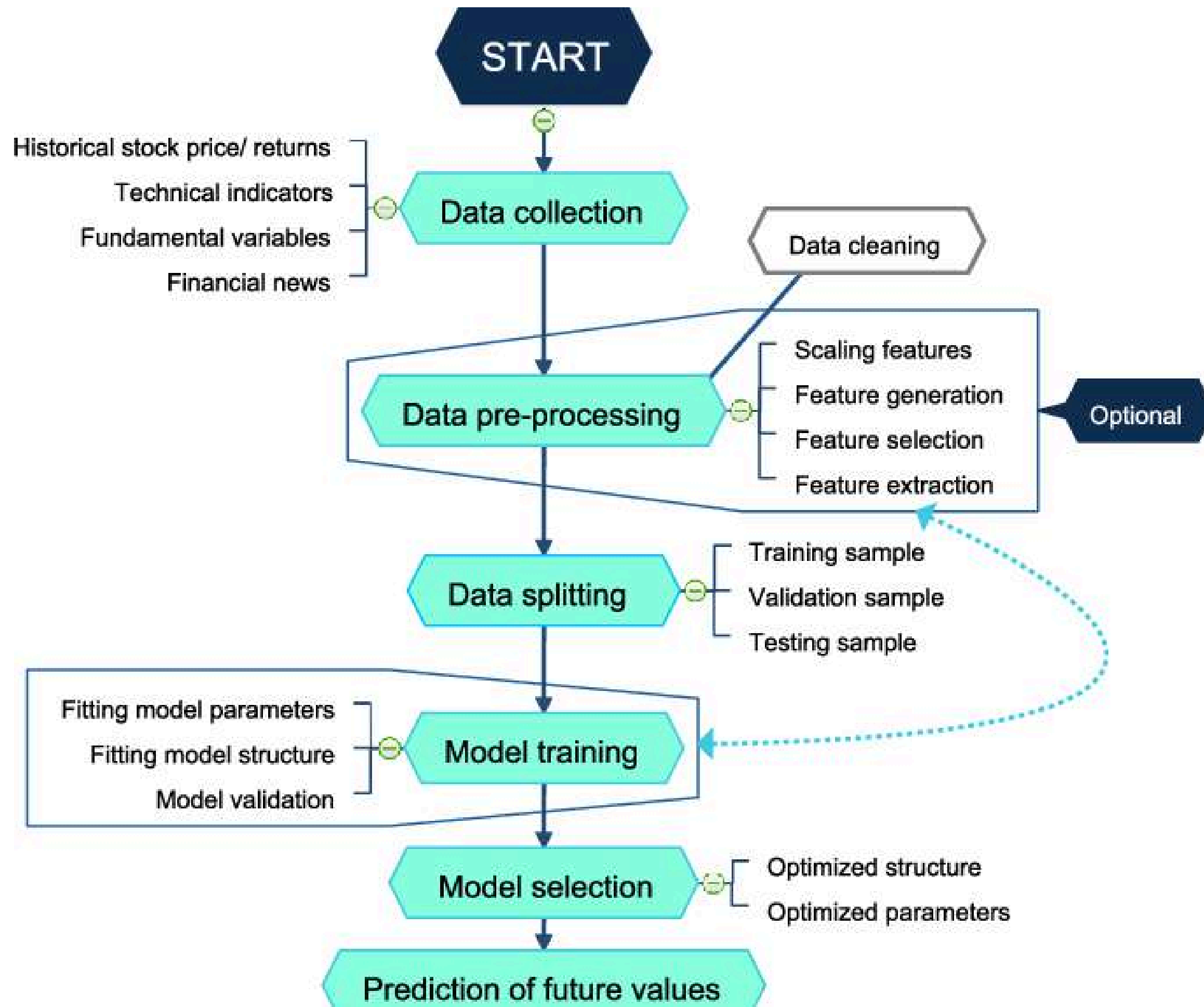


Objectives:

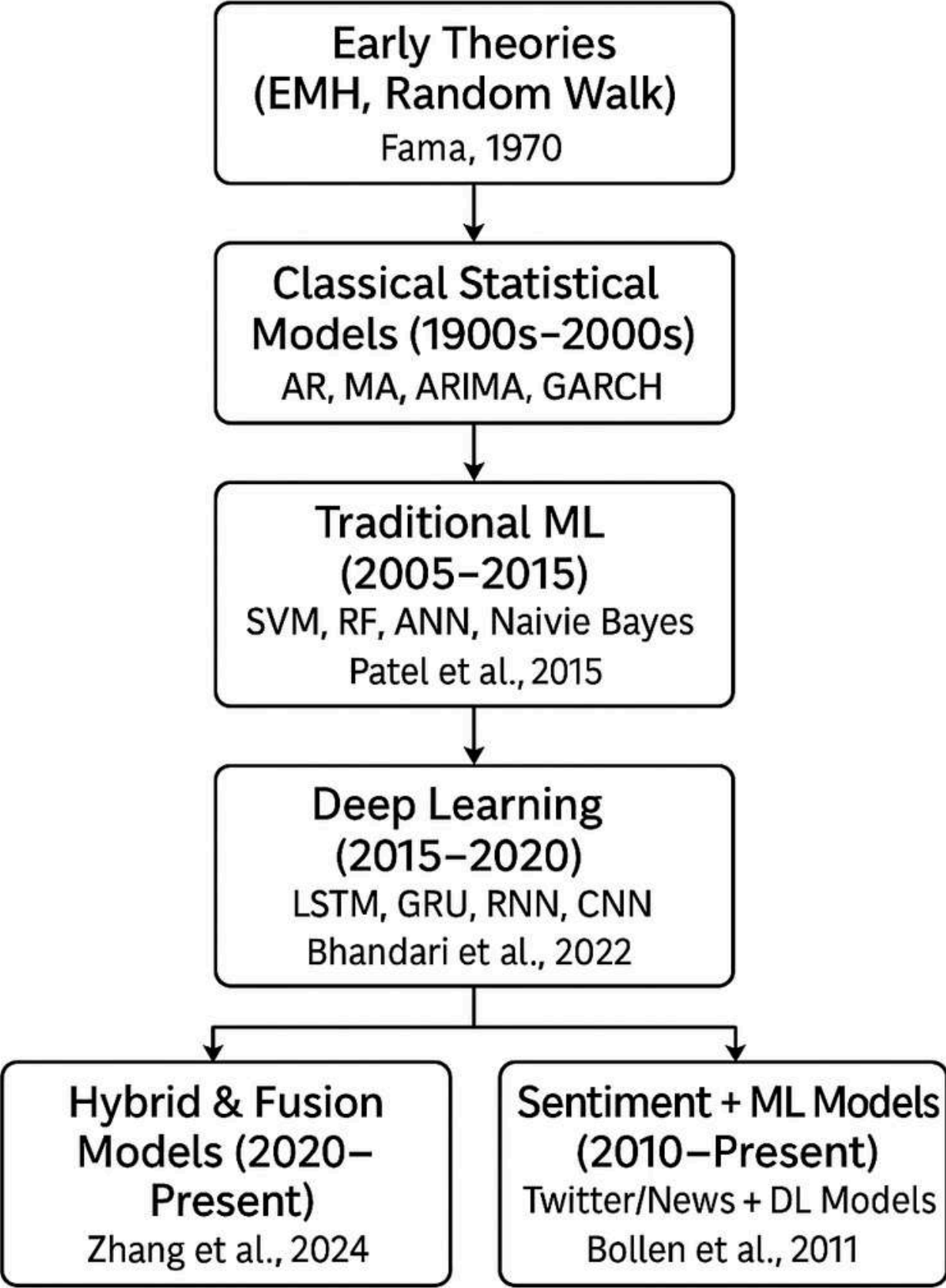
- **Provides almost accurate stock price predictions**
- **Helps investors identify buy/sell opportunities**
- **Delivers real-time market insights**
- **Provides news about recent trends in stock market**



Workflow:



Evolution of ML Models for Stock Market Prediction

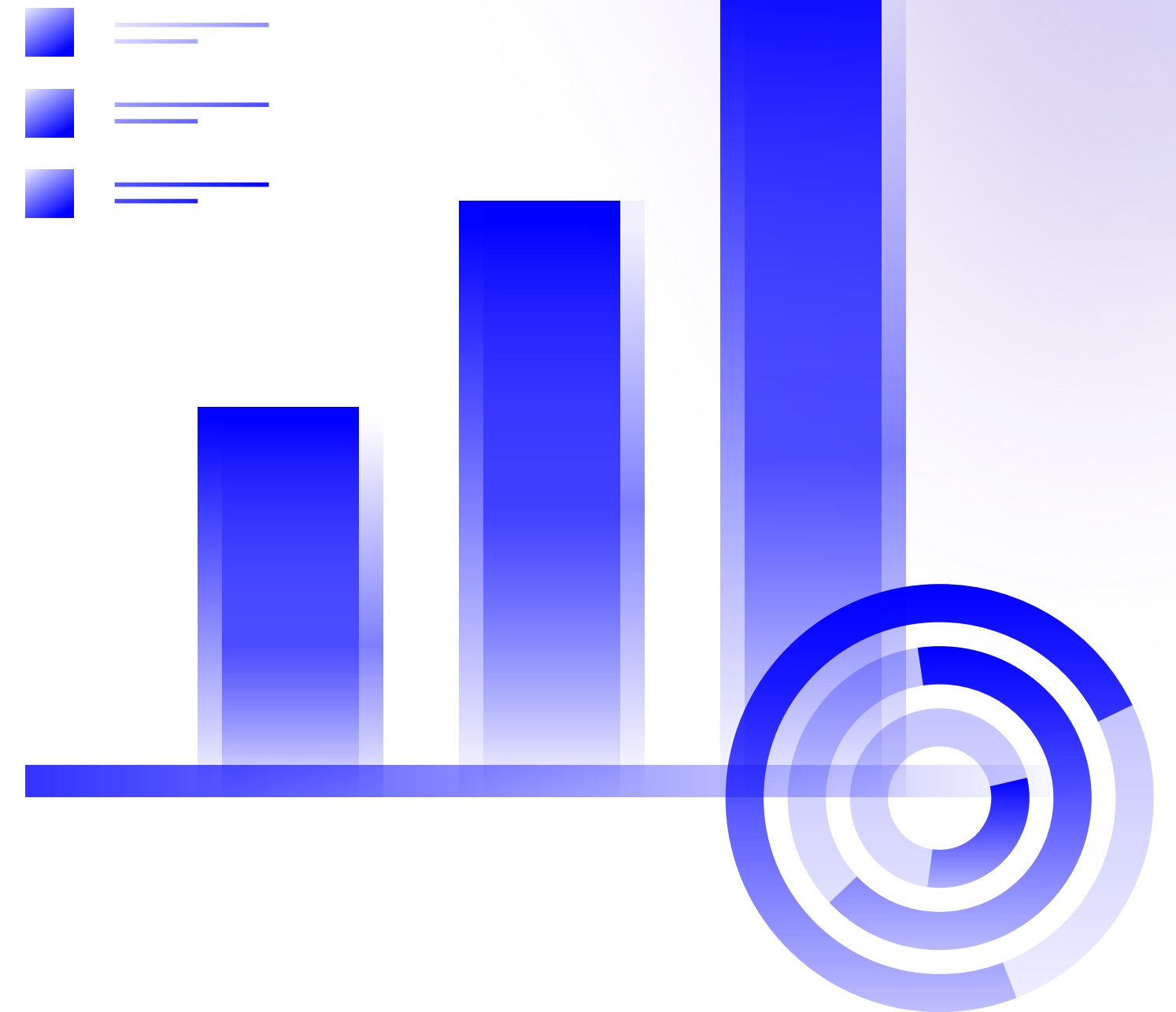


Model	Advantages	Disadvantages
ARIMA	Simple, interpretable, good for linear short-term patterns	Poor for nonlinear data, requires stationarity
GARCH	Best for volatility modeling	Predicts variance, not price; limited nonlinear ability
SVM	Good for nonlinear patterns, works with small data	Not sequential; expensive on large data
LSTM	Excellent for long-term dependencies; captures nonlinear patterns	Slow training; needs large data
GRU	Faster than LSTM and good temporal modeling	Slightly less powerful; still heavy computationally

Model Selection: LSTM 64 + Dropout + LSTM 32

Stocks used for model selection

- **GOOGL**
- **MSFT**
- **AAPL**
- **IBM**
- **TSLA**



GOOGL

=====

REAL-WORLD PERFORMANCE

=====

Next-Day Directional Accuracy: 51.95%

Classification Matrix Metrics:

	precision	recall	f1-score	support
Down/Flat	0.40	0.04	0.08	241
Up	0.53	0.94	0.68	271
accuracy			0.52	512
macro avg	0.46	0.49	0.38	512
weighted avg	0.47	0.52	0.39	512

=====

LIVE NEXT-DAY FORECAST FOR THE NEXT SESSION

=====

Probability that GOOGL will finish HIGHER tomorrow: 52.80%

Actionable Model Decision: BUY / LONG

=====

=====

PREDICTED NEXT-DAY STOCK PRICE

=====

Last Known Close (2026-06-30): \$357.37

Predicted Direction: UP (Model Confidence: 52.80%)

Estimated Next-Day Closing Price: \$361.84 (implied move: +1.25%)

Model's Historical Directional Accuracy on Test Set: 51.95%



Fig 1. GOOGL Price Prediction Graph

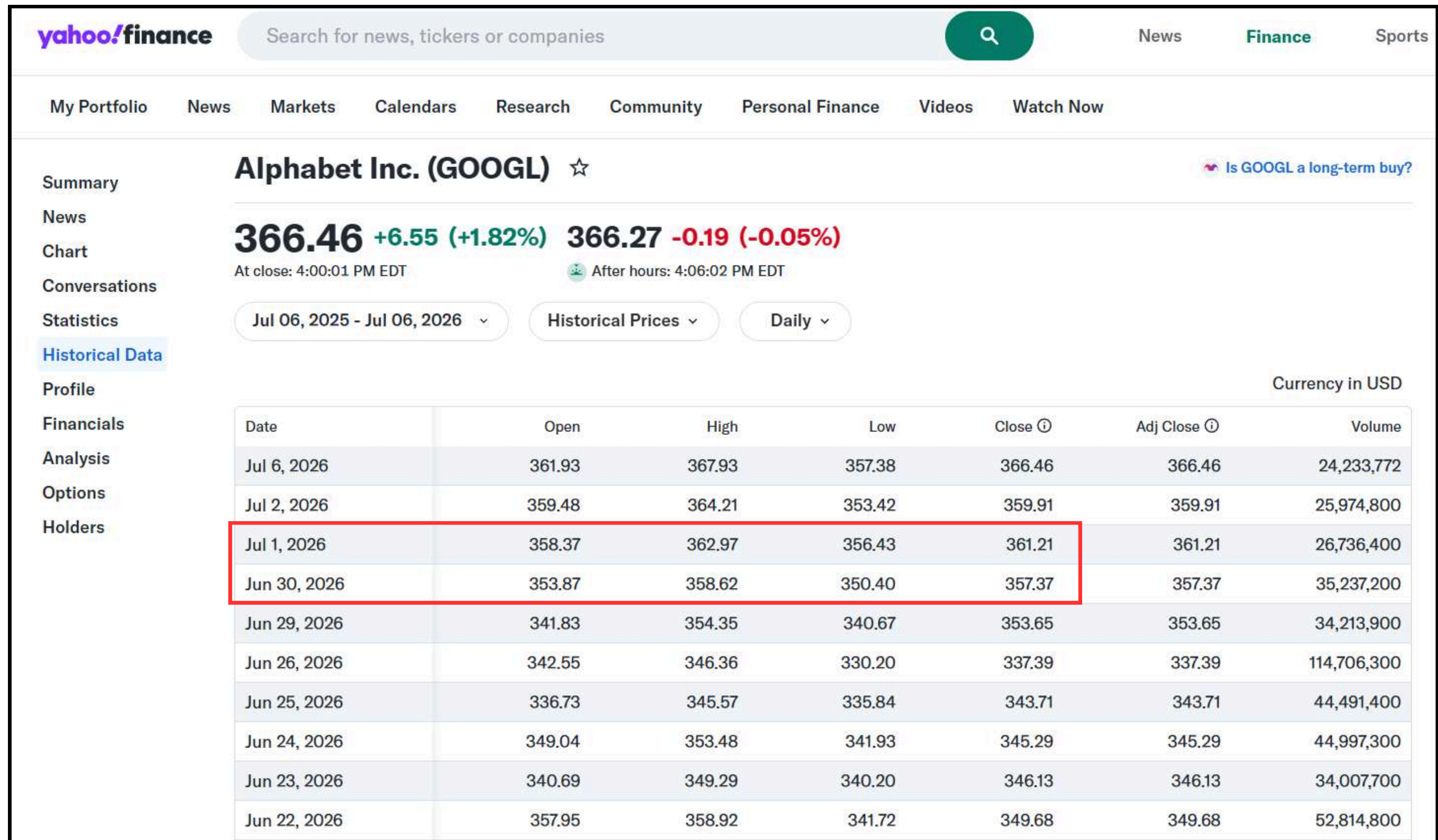


Fig 2. GOOGL Historical Data

MSFT

=====

REAL-WORLD PERFORMANCE

=====

Next-Day Directional Accuracy: 50.00%

Classification Matrix Metrics:

	precision	recall	f1-score	support
Down/Flat	0.44	0.14	0.21	247
Up	0.51	0.84	0.63	265
accuracy			0.50	512
macro avg	0.48	0.49	0.42	512
weighted avg	0.48	0.50	0.43	512

=====

LIVE NEXT-DAY FORECAST FOR THE NEXT SESSION

=====

Probability that MSFT will finish HIGHER tomorrow: 56.61%

Actionable Model Decision: BUY / LONG

=====

=====

PREDICTED NEXT-DAY STOCK PRICE

=====

Last Known Close (2026-06-30): \$373.02

Predicted Direction: UP (Model Confidence: 56.61%)

Estimated Next-Day Closing Price: \$377.60 (implied move: +1.23%)

Model's Historical Directional Accuracy on Test Set: 50.00%

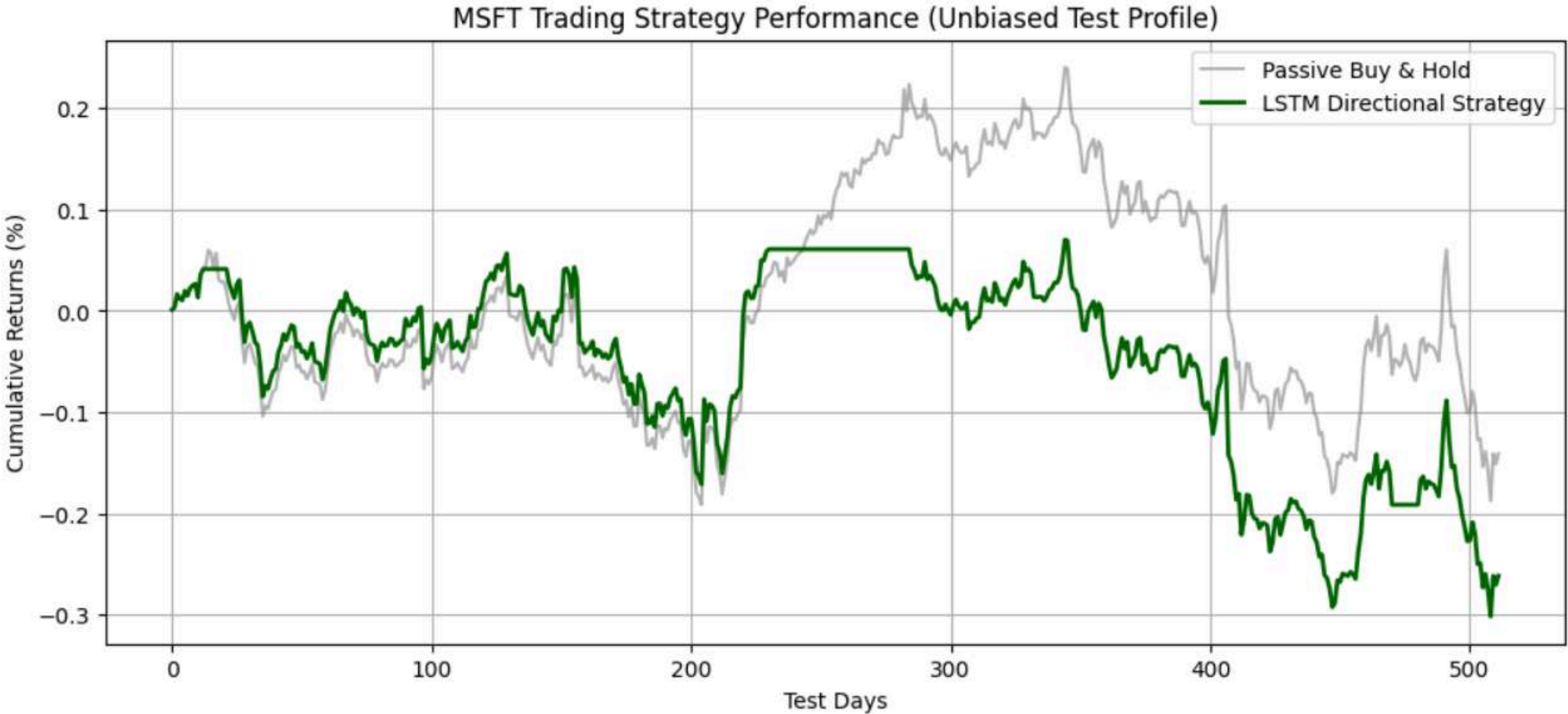


Fig 3. MSFT Price Prediction Graph

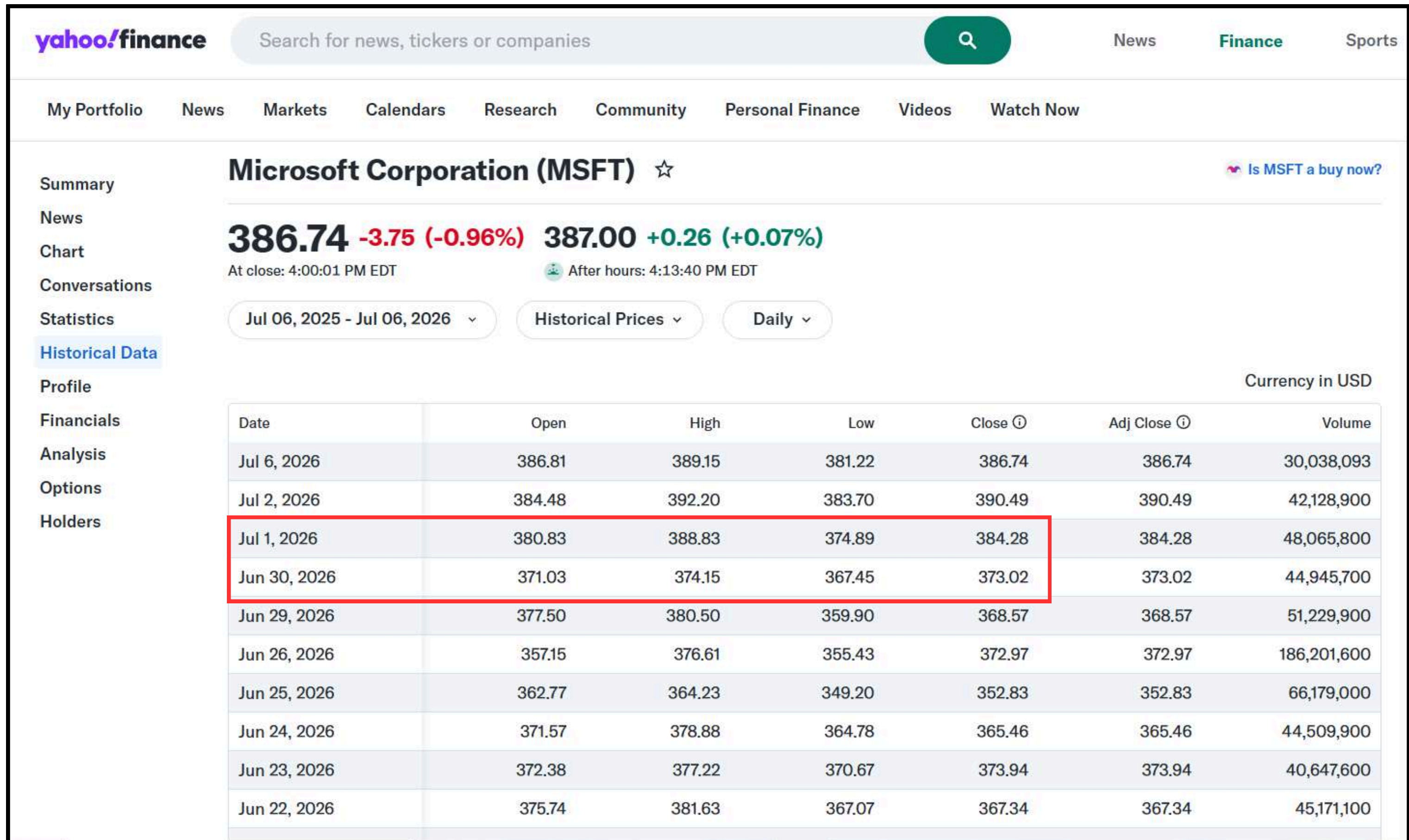


Fig 4. MSFT Historical Data

AAPL

=====

REAL-WORLD PERFORMANCE

=====

Next-Day Directional Accuracy: 53.12%

Classification Matrix Metrics:

	precision	recall	f1-score	support
Down/Flat	0.49	0.64	0.56	233
Up	0.60	0.44	0.50	279
accuracy			0.53	512
macro avg	0.54	0.54	0.53	512
weighted avg	0.55	0.53	0.53	512

=====

LIVE NEXT-DAY FORECAST FOR THE NEXT SESSION

=====

Probability that AAPL will finish HIGHER tomorrow: 43.40%

Actionable Model Decision: SELL / SHORT or CASH

=====

=====

PREDICTED NEXT-DAY STOCK PRICE

=====

Last Known Close (2026-06-30): \$289.36

Predicted Direction: DOWN (Model Confidence: 43.40%)

Estimated Next-Day Closing Price: \$285.79 (implied move: -1.24%)

Model's Historical Directional Accuracy on Test Set: 53.12%

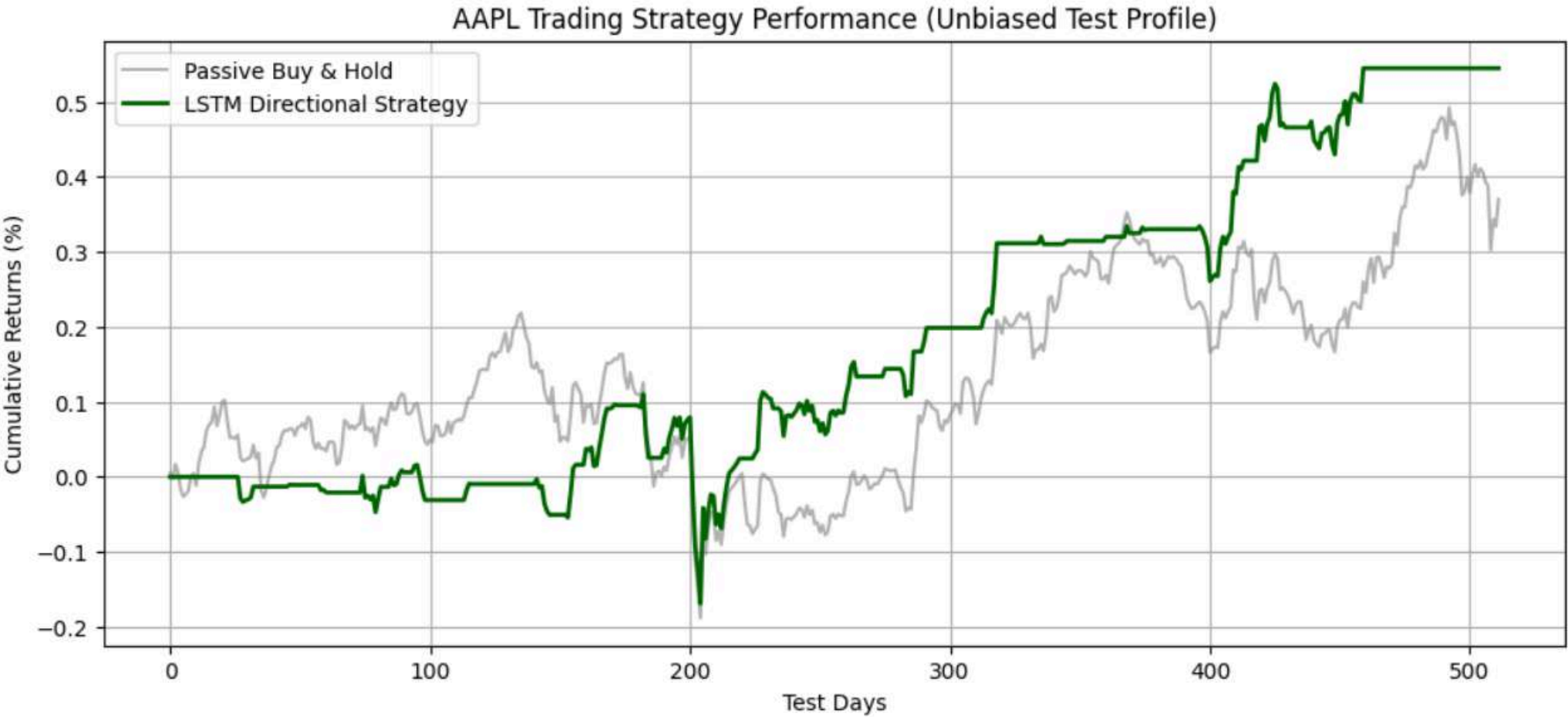


Fig 5. AAPL Price Prediction Graph

TSLA

=====

REAL-WORLD PERFORMANCE

=====

Next-Day Directional Accuracy: 52.34%

Classification Matrix Metrics:

	precision	recall	f1-score	support
Down/Flat	0.50	0.42	0.46	245
Up	0.54	0.62	0.57	267
accuracy			0.52	512
macro avg	0.52	0.52	0.52	512
weighted avg	0.52	0.52	0.52	512

=====

LIVE NEXT-DAY FORECAST FOR THE NEXT SESSION

=====

Probability that TSLA will finish HIGHER tomorrow: 42.34%

Actionable Model Decision: SELL / SHORT or CASH

=====

=====

PREDICTED NEXT-DAY STOCK PRICE

=====

Last Known Close (2026-06-30): \$420.60

Predicted Direction: DOWN (Model Confidence: 42.34%)

Estimated Next-Day Closing Price: \$410.52 (implied move: -2.40%)

Model's Historical Directional Accuracy on Test Set: 52.34%

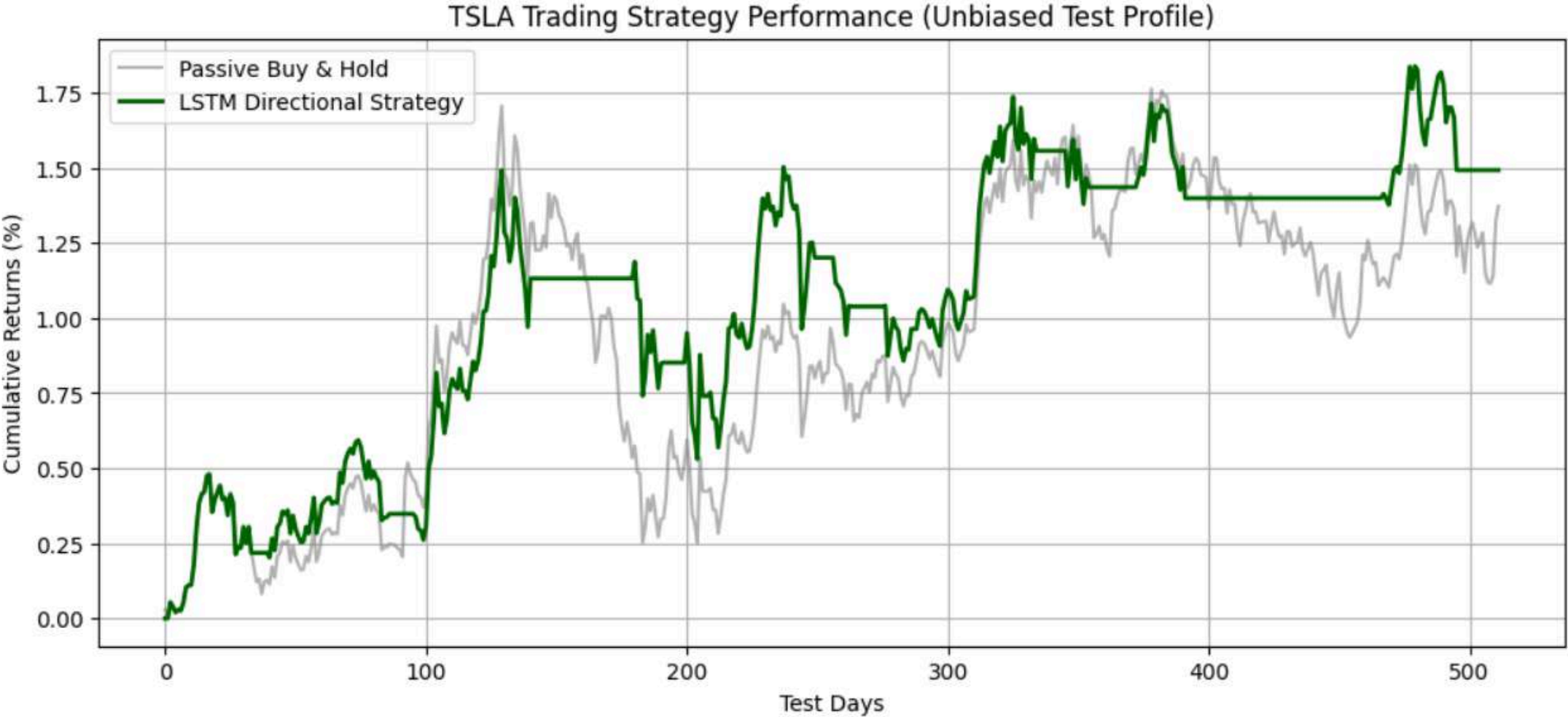


Fig 6. TSLA Price Prediction Graph

IBM

REAL-WORLD PERFORMANCE

Next-Day Directional Accuracy: 52.34%

Classification Matrix Metrics:

	precision	recall	f1-score	support
Down/Flat	0.46	0.44	0.45	225
Up	0.57	0.59	0.58	287
accuracy			0.52	512
macro avg	0.51	0.51	0.51	512
weighted avg	0.52	0.52	0.52	512

LIVE NEXT-DAY FORECAST FOR THE NEXT SESSION

Probability that IBM will finish HIGHER tomorrow: 47.67%
Actionable Model Decision: SELL / SHORT or CASH

PREDICTED NEXT-DAY STOCK PRICE

Last Known Close (2026-07-02): \$289.52
Predicted Direction: DOWN (Model Confidence: 47.67%)
Estimated Next-Day Closing Price: \$286.56 (implied move: -1.02%)
Model's Historical Directional Accuracy on Test Set: 52.34%

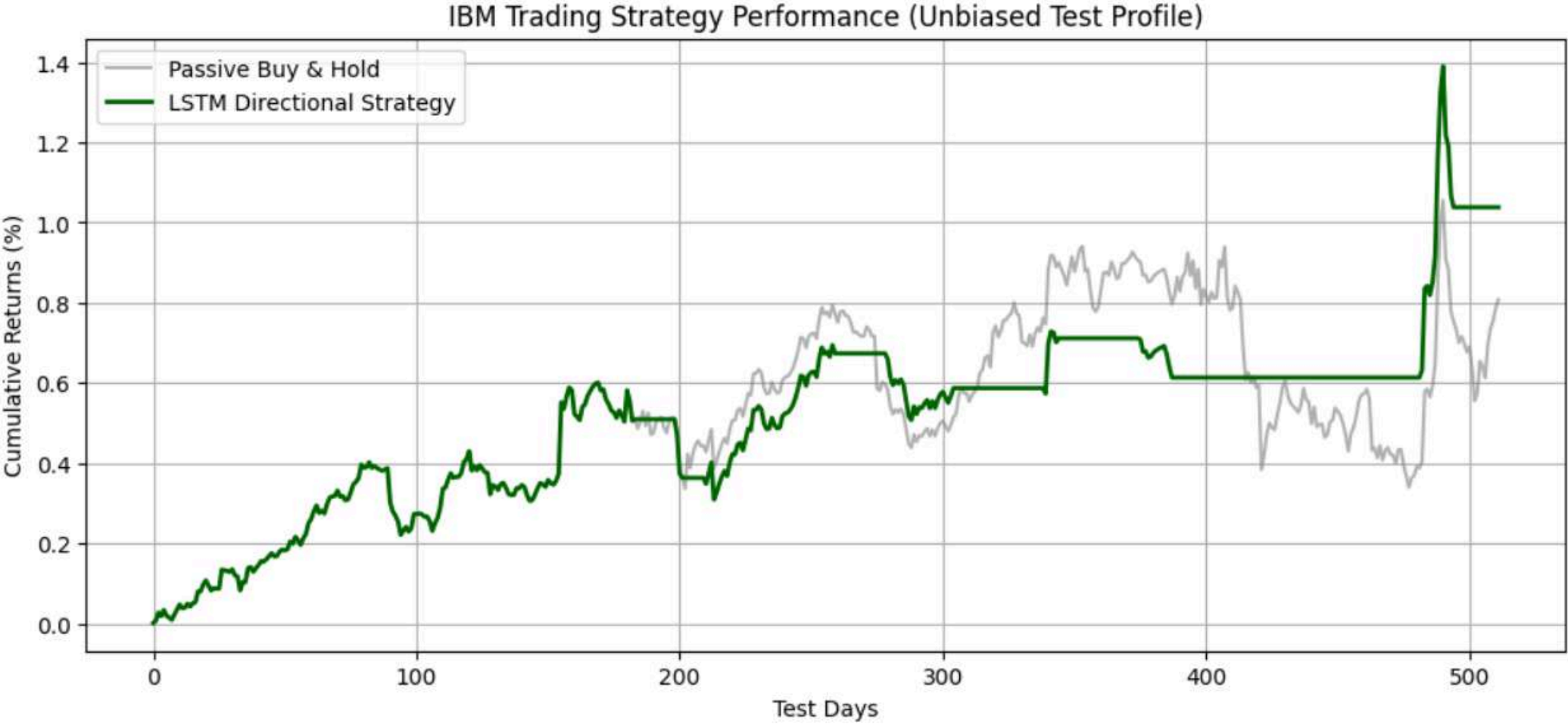
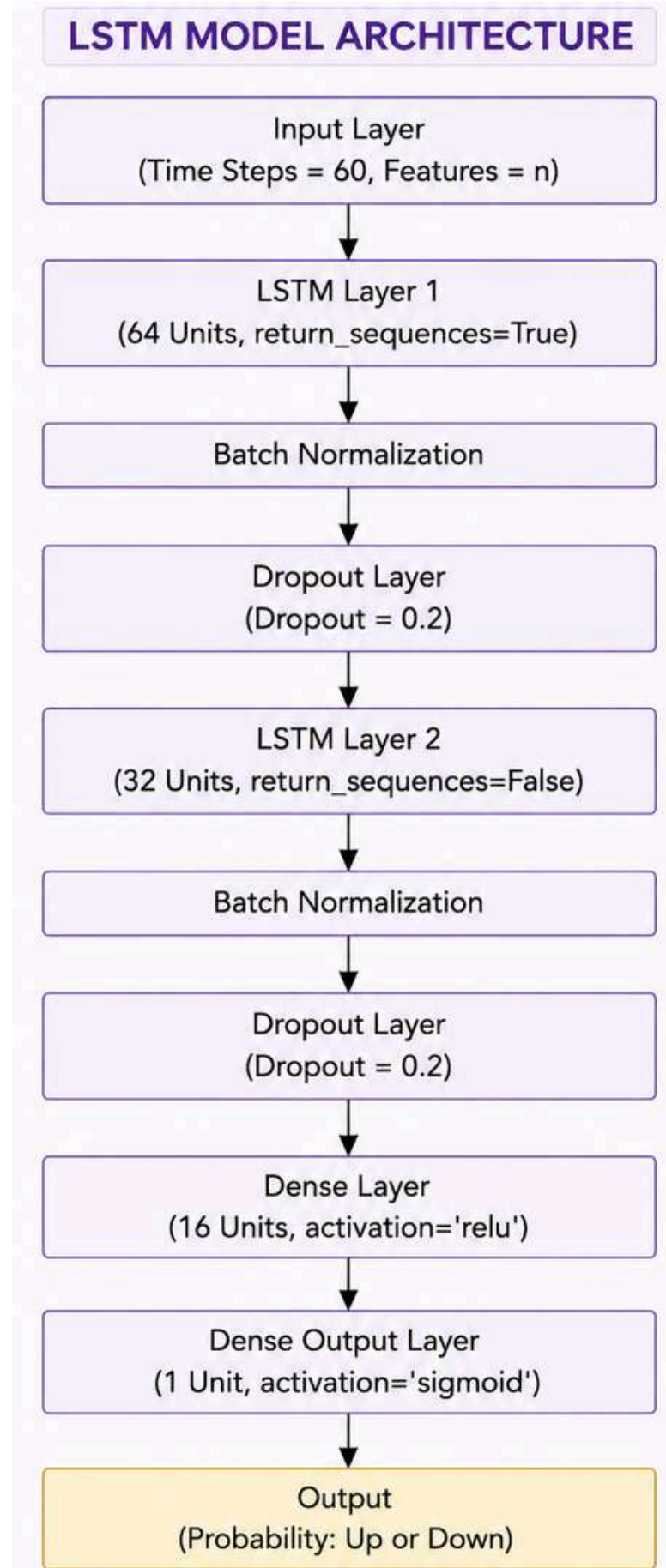


Fig 7. IBM Price Prediction Graph

Model Architecture:



Why LSTM 64 + Dropout + LSTM 32 is selected?

- **Dual-Layer Deep Learning:** Uses stacked LSTMs (64+32 units) to separate high-level market macro-trends from immediate, short-term price reactions.
- **Noise Filtering:** Features a 20% Dropout layer that prevents the model from "memorizing" erratic market noise, ensuring highly generalized predictions.
- **Profit-Driven Loss Function:** Implements a custom Directional Loss that applies a 2 X penalty for wrong direction guesses, prioritizing directional accuracy over mere price proximity.
- **Structural Synergy:** Avoids hybrid (LSTM+GRU) bottlenecks by using an all-LSTM pipeline, ensuring smooth gradient flow and faster convergence.

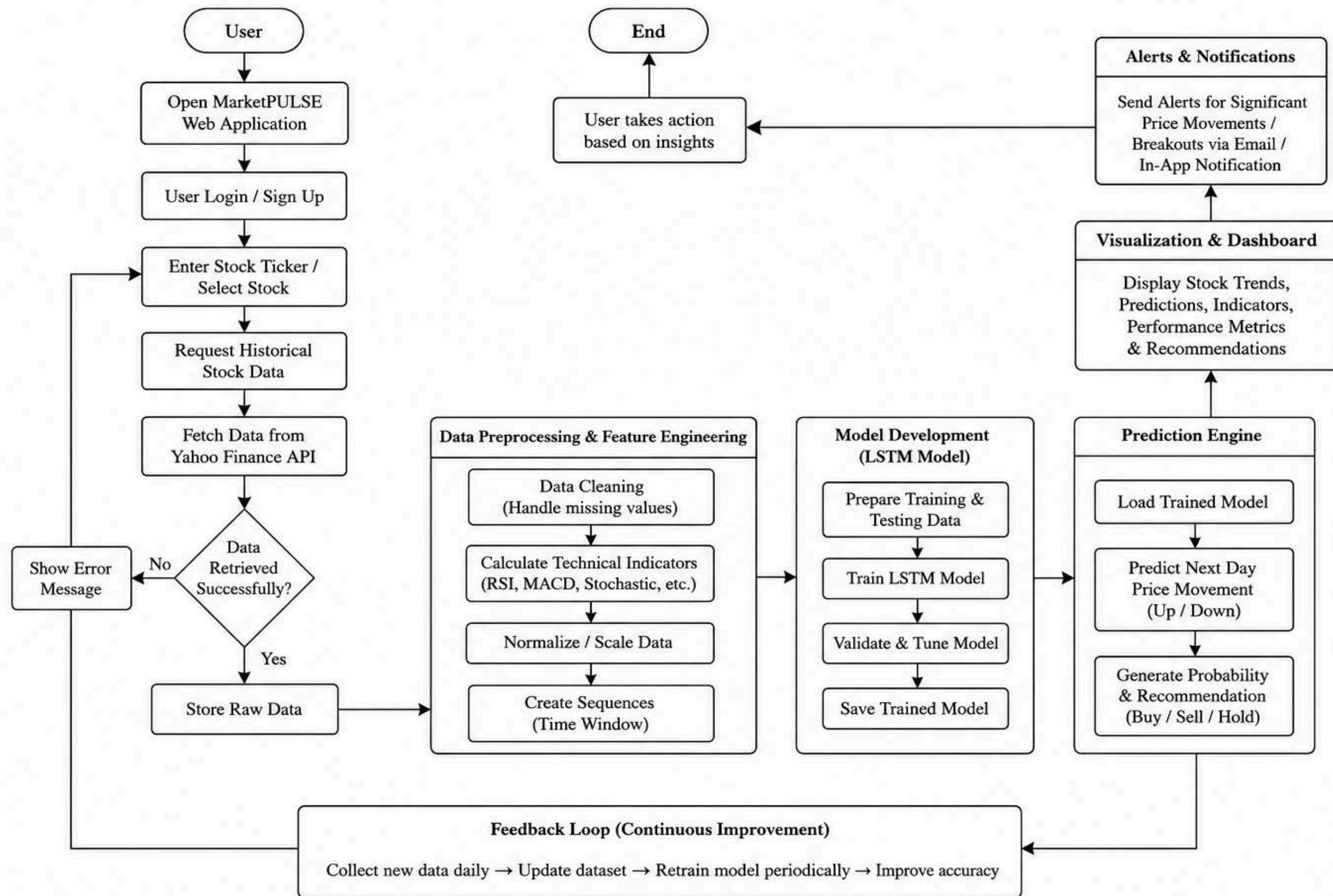


Fig 8. Flowchart of MarketPULSE

Technical Requirements:

- **Frontend:** HTML, CSS, JS
 - **Backend:** Python (TensorFlow, Keras, Pandas, NumPy, Scikit-Learn, ta, Matplotlib)
 - **Web Framework:** Flask
 - **IDE:** Visual Studio Code
 - **Model Development, Training, Testing:** Google Colab
 - **Operation System:** Windows 10/11, Linux, or macOS
 - **Processor:** Intel Core i5 or higher
 - **RAM:** Minimum 8 GB
 - **Web Browser:** Google Chrome, Microsoft Edge, Mozilla Firefox
- 

Methodology and Implementation:

Presentation Layer (HTML, CSS, JS)

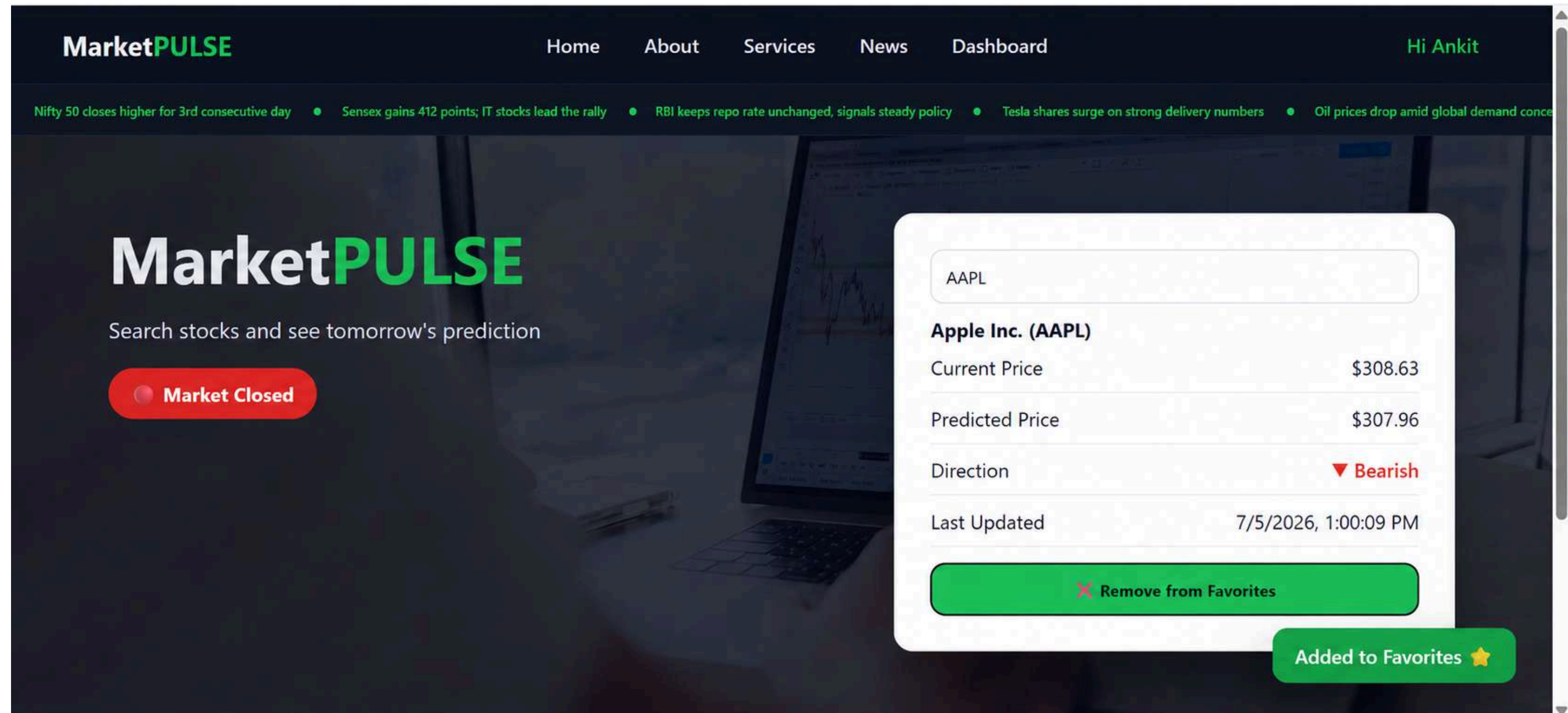


Fig 9. Landing Page

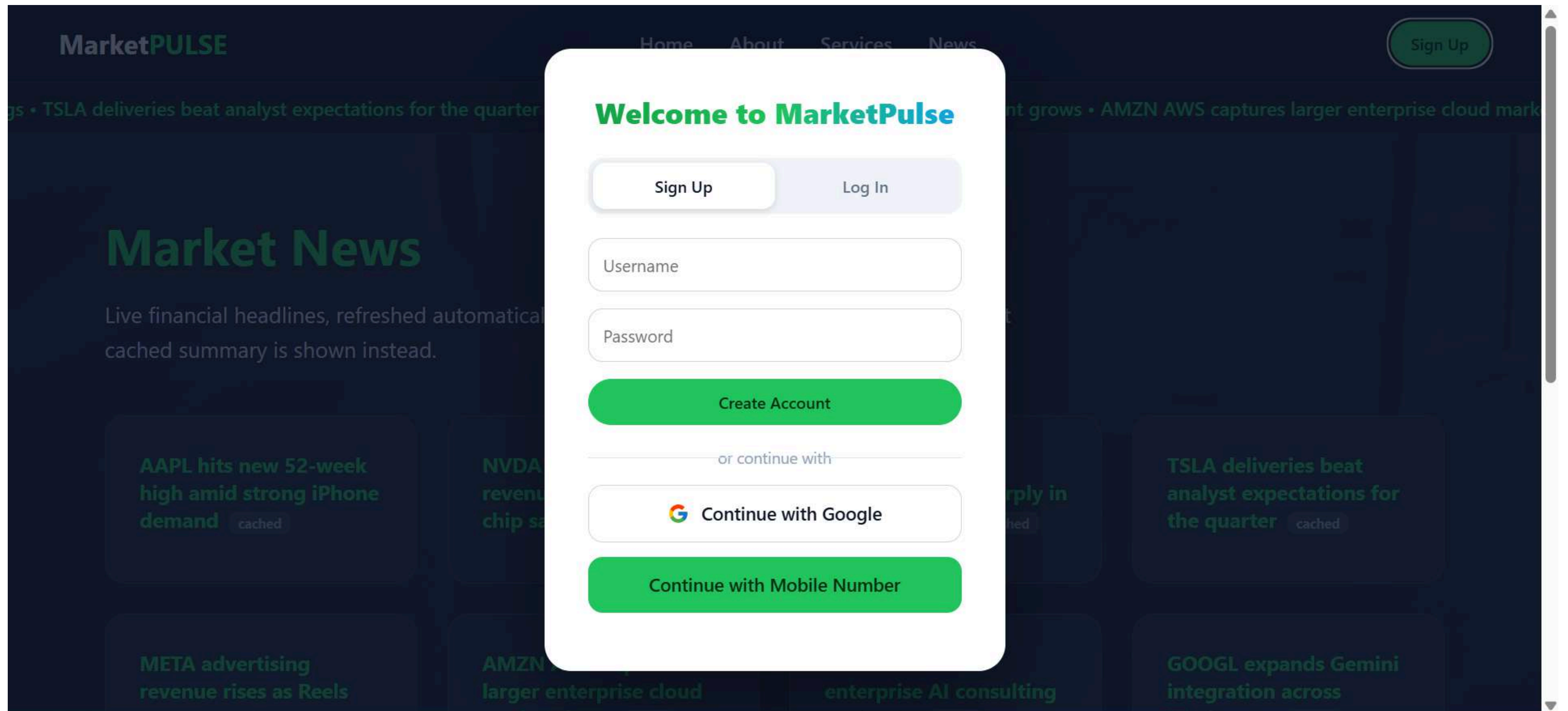


Fig 10. User Login Page

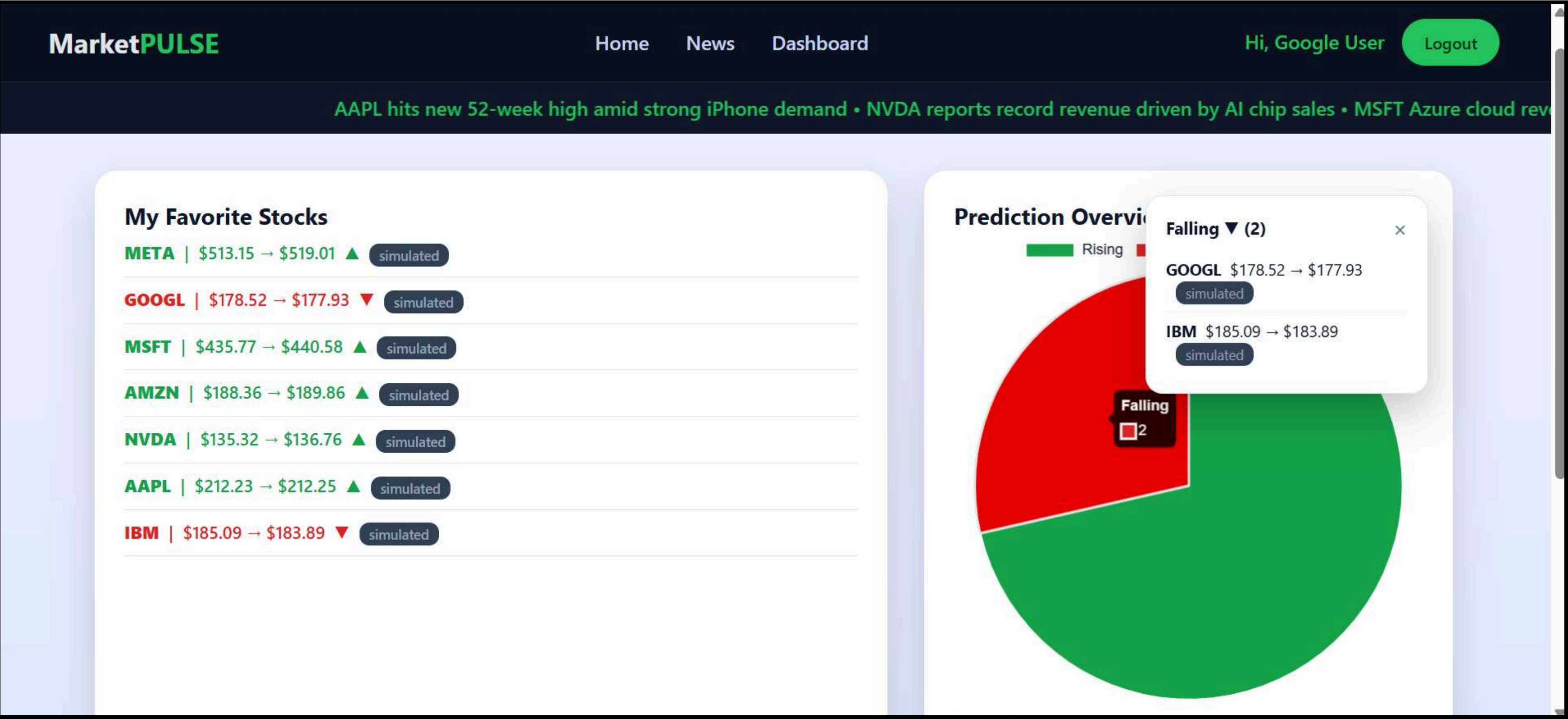


Fig 11. User Dashboard Page

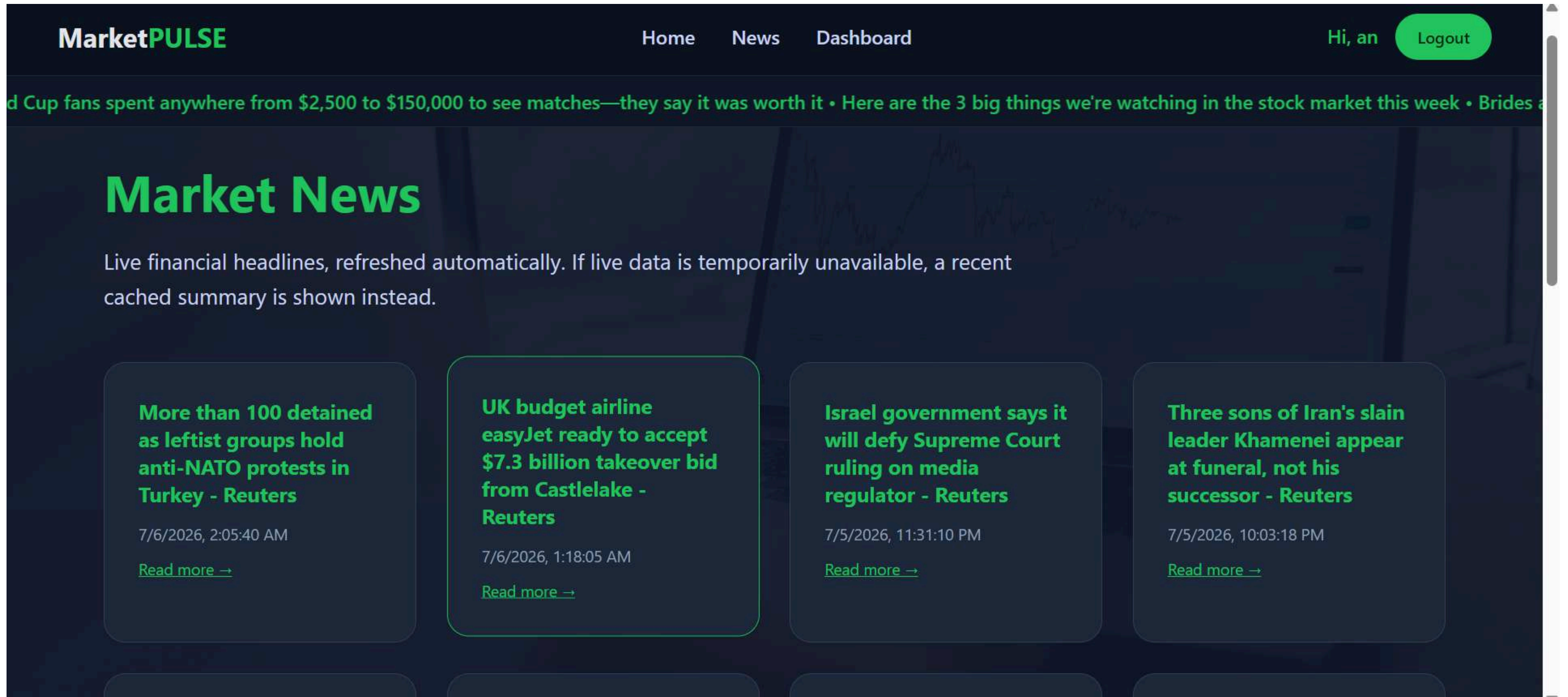


Fig 12. Live News Feed Page

Application Layer (Python)

```
!pip install yfinance ta -q

import numpy as np
import pandas as pd
import yfinance as yf
import matplotlib.pyplot as plt

from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout, BatchNormalization
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
from tensorflow.keras.optimizers import Adam
|
from ta.momentum import RSIIndicator, StochasticOscillator
from ta.trend import MACD
from datetime import datetime

TICKER = "TSLA"
START = "2015-01-01"
#END = datetime.today().strftime('%Y-%m-%d')
END = "2026-07-01"
SEQ_LENGTH = 60
```

Fig 12. Code Snippet

Data Layer (API Sources- YahooFinance)

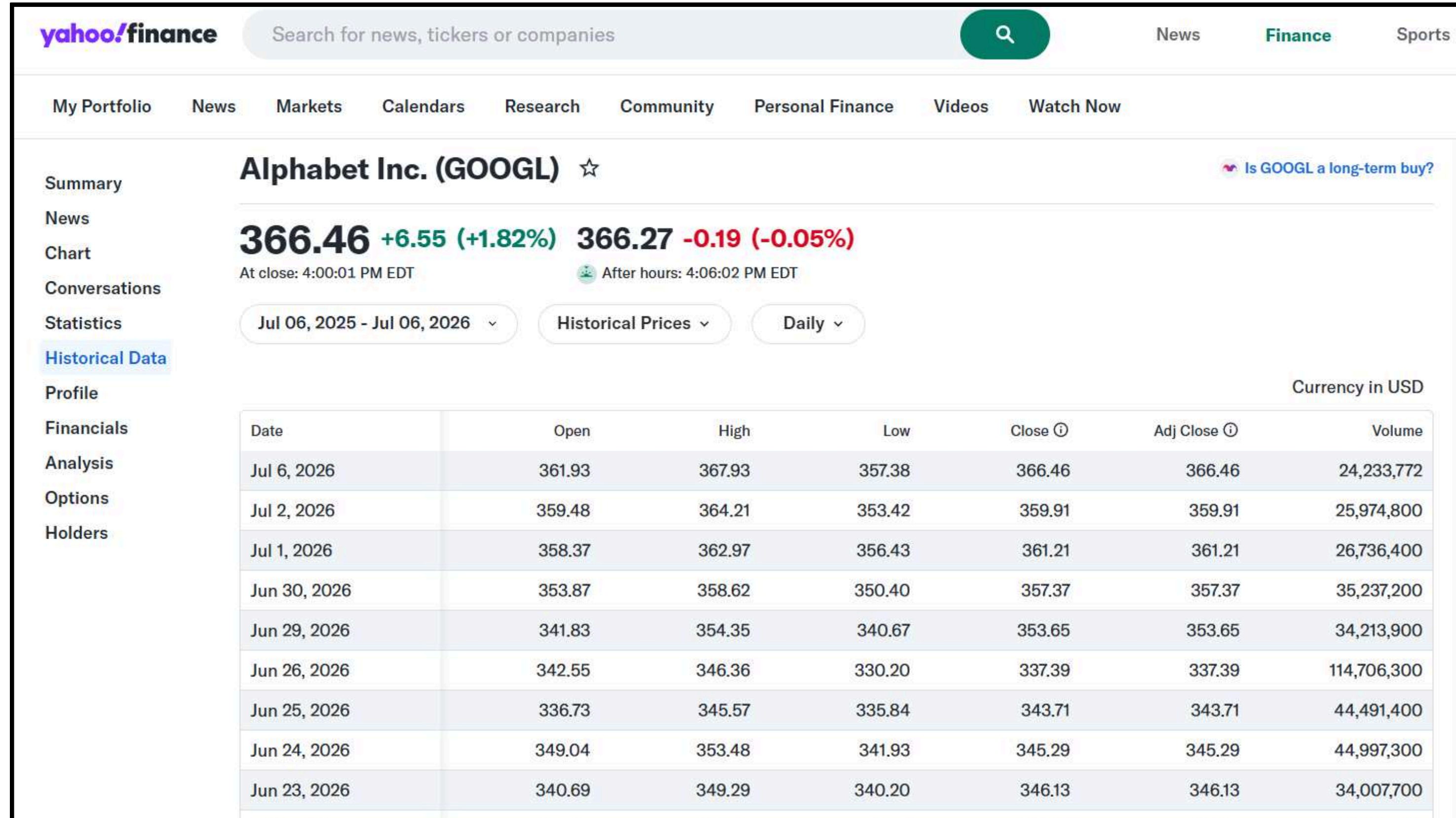


Fig 13. Historical Data

MarketPULSE: Real Time Stock Insights, Trends & Alerts

Project Gantt Chart

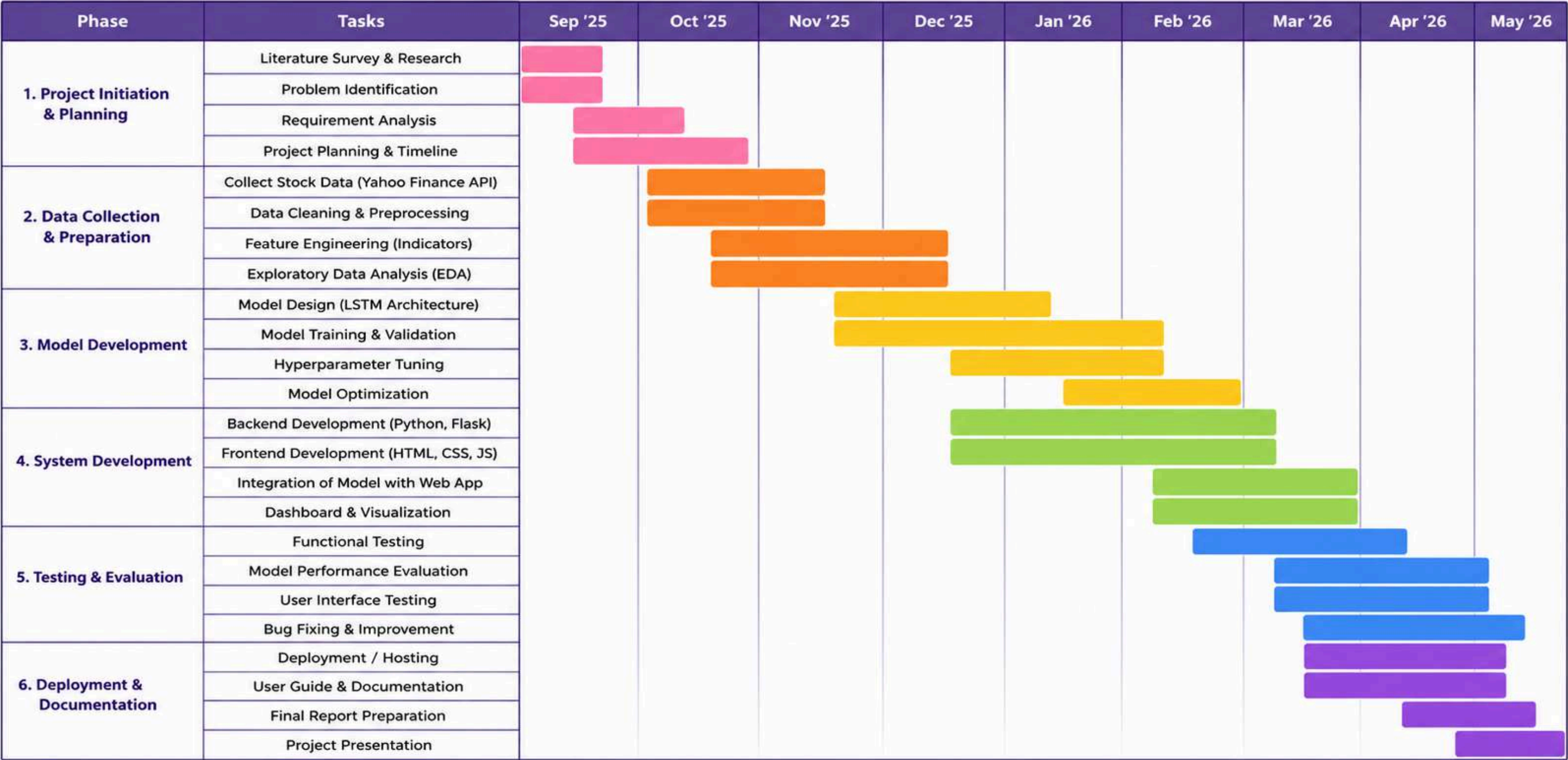


Fig 14. Gantt Chart

Result:

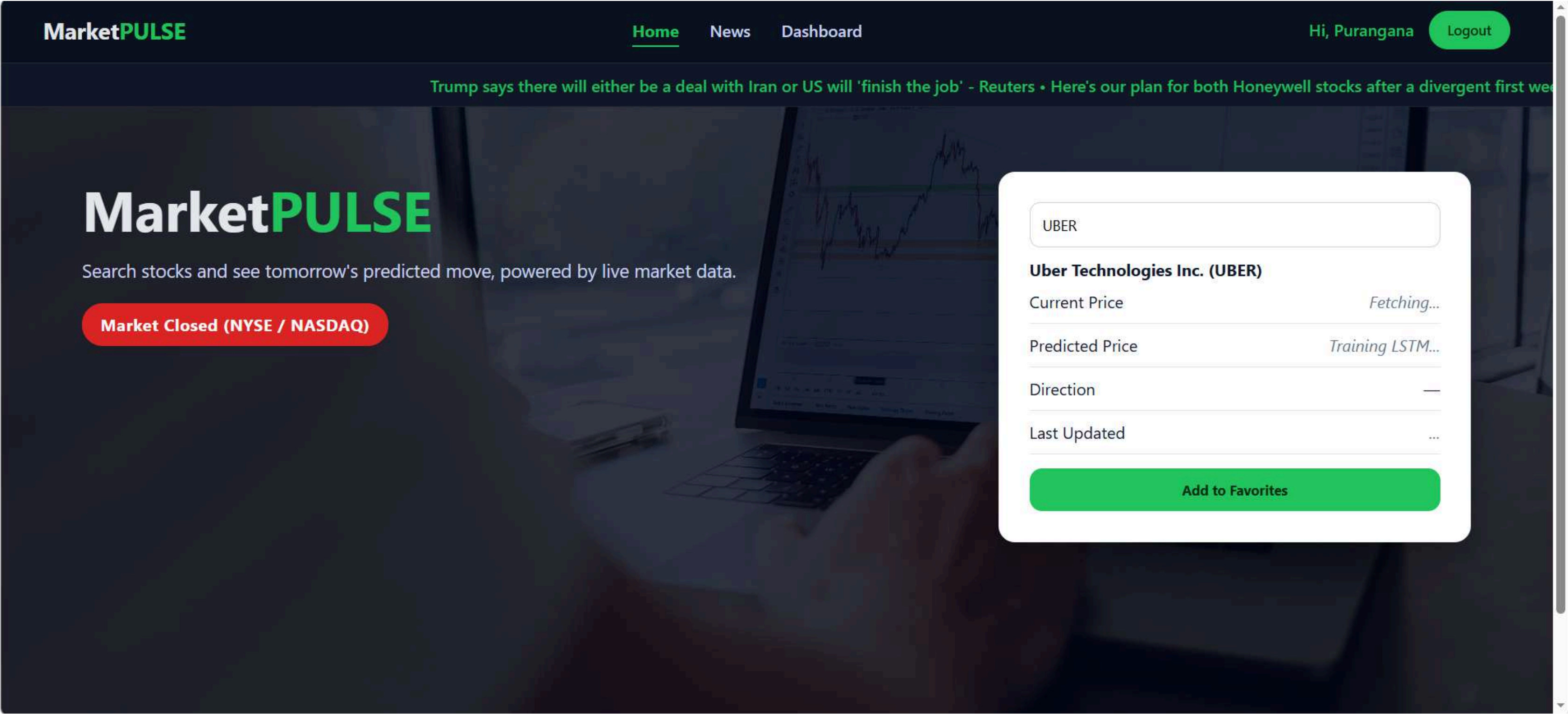


Fig 15. Live Data Fetching

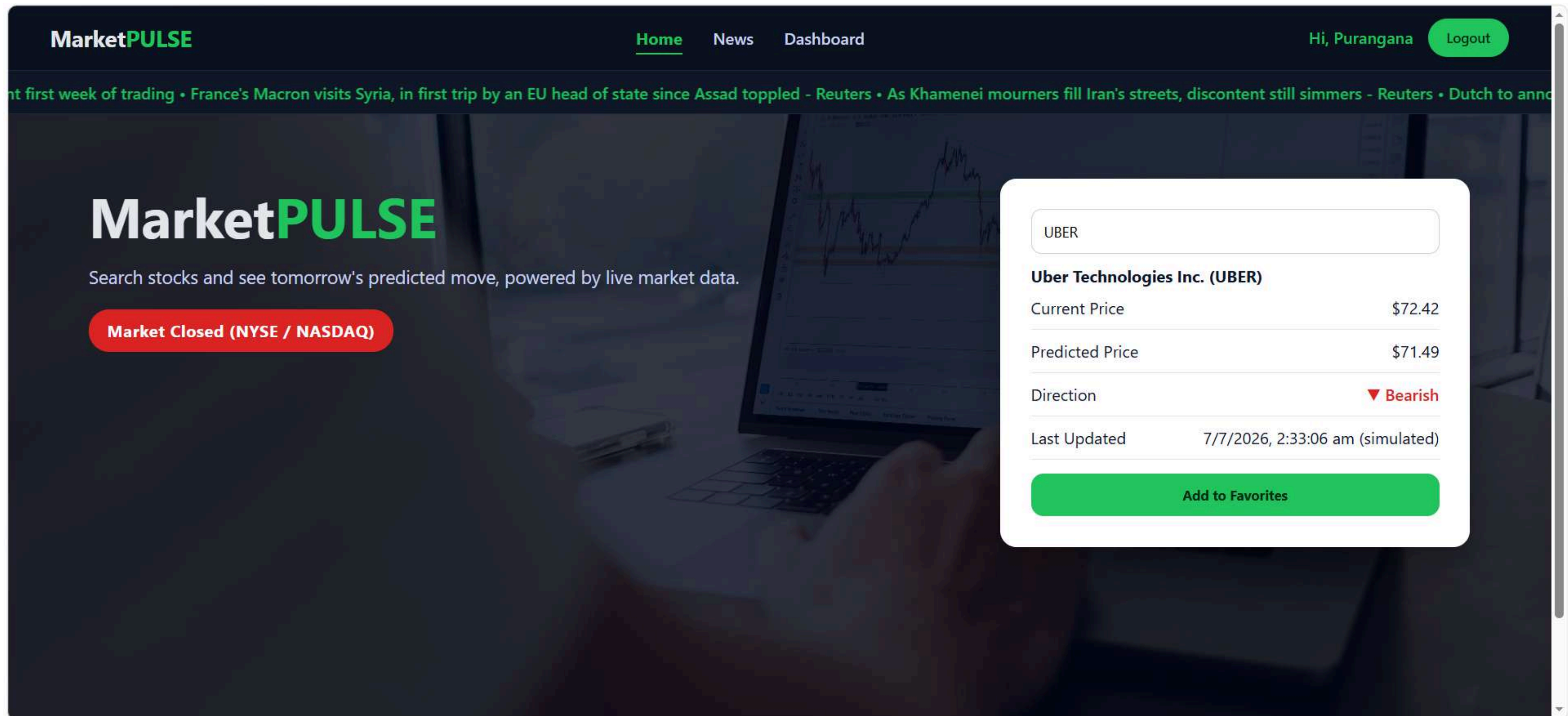


Fig 16. Predicted Price

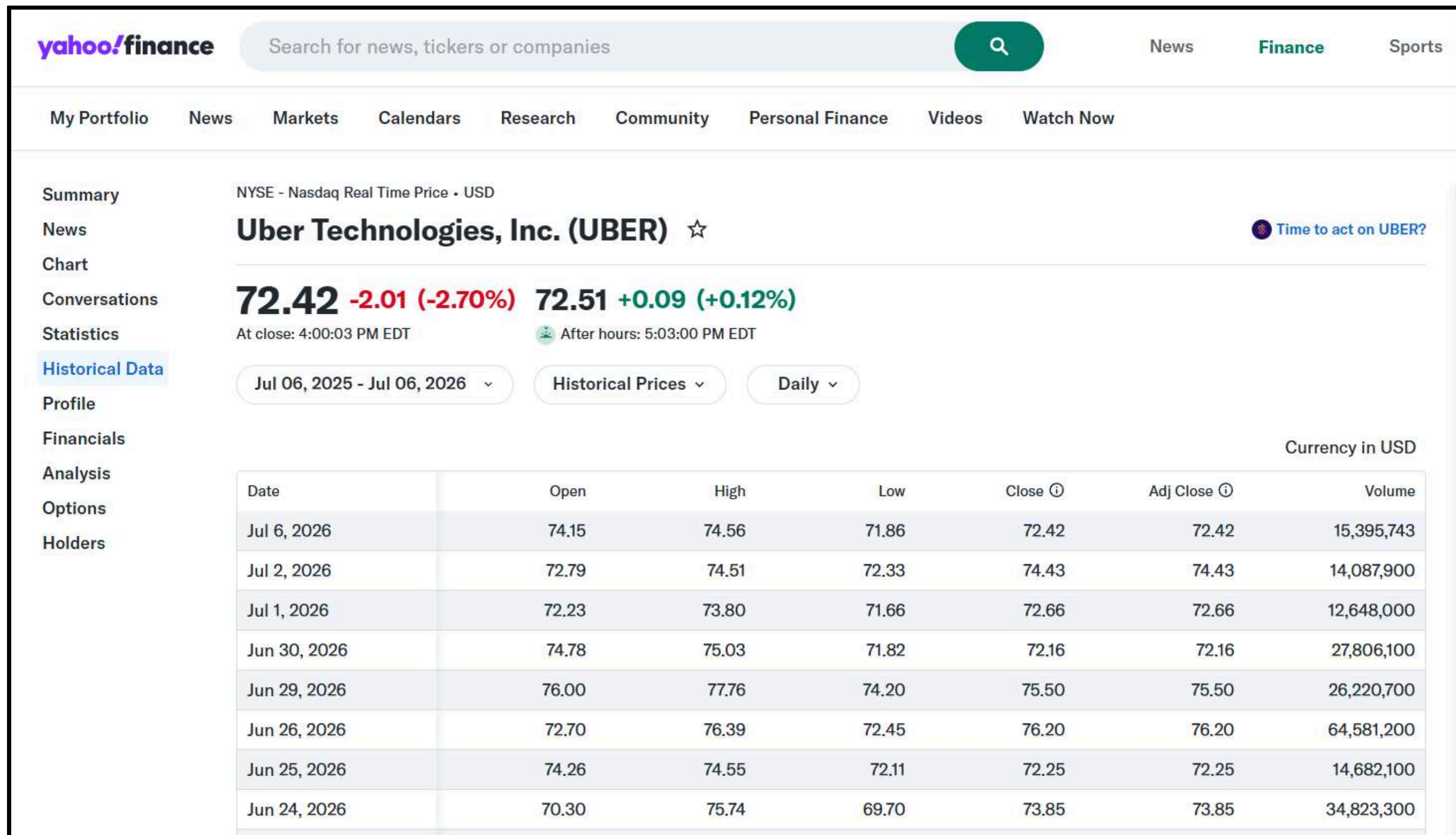
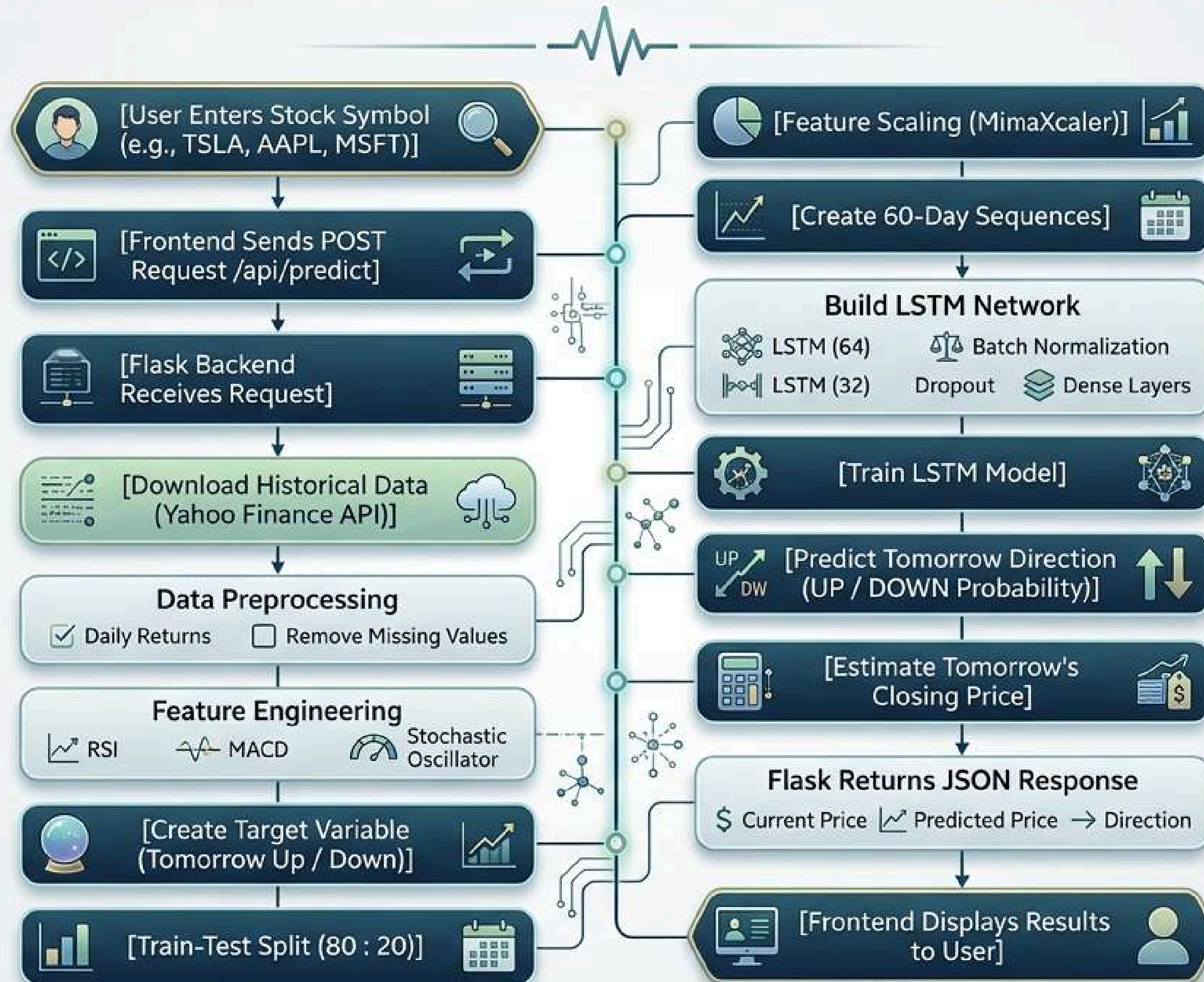


Fig 17. Historical Data

MARKETPULSE PREDICTION WORKFLOW



Challenges:

- **Handling unpredictable market events (e.g., global crises).**
- **Maintaining data accuracy & reducing noise.**
- **Avoiding overfitting in ML models.**
- **Managing latency for real-time predictions.**





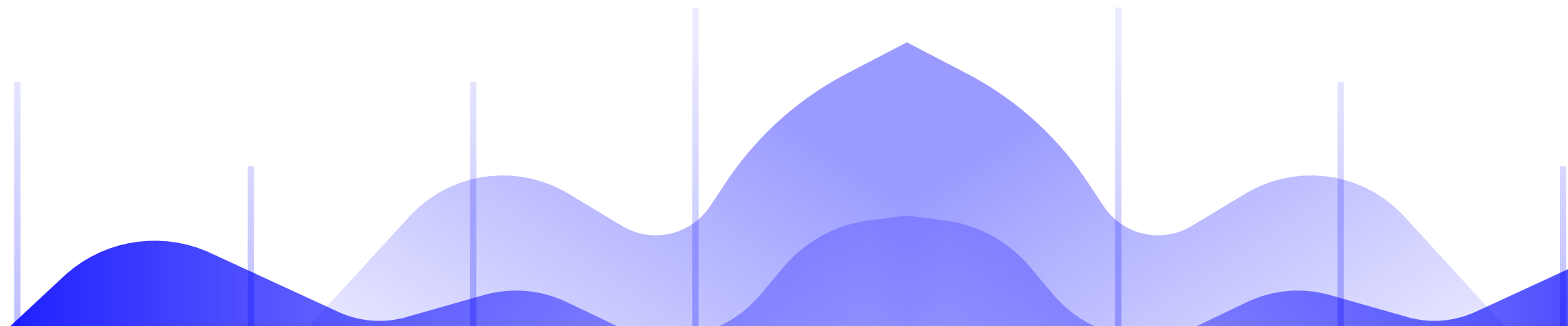
Performance:

- **Improved Prediction Accuracy**
- **Enhanced Trend Detection and Analysis**
- **Highly Effective Real-Time Alerts**
- **High Scalability and Real-Time System Performance**



Strengths of MarketPULSE (USPs):

- **Multi-Feature Integration:** Combines prediction, news, alerts, and suggestions in one app.
- **Real-Time Insights:** Faster updates compared to traditional apps.
- **AI-Powered Predictions:** Uses deep learning for higher accuracy.
- **User-Centric Design:** Simple, interactive dashboards for all levels of investors.
- **Custom Alerts:** Personalized notifications for favourite stocks.
- **Balanced Approach:** Covers both technical (charts, predictions) and fundamental (news, trends) analysis.



References:

- [1] Fama, Eugene Francis. (1970). Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance*, 25(2), 383–417.
- [2] Patel, Jigar, Shah, Sahil, Thakkar, Priyank, & Kotecha, Ketan. (2015). Predicting Stock Market Index Using Fusion of Machine Learning Techniques. *Expert Systems with Applications*, 42(4), 2162–2172.
- [3] Bhandari, Harsh Narayan, Rimal, Bishal Prasad, Pokharel, Saugat, and Ghimire, Ashok. (2022). Predicting Stock Market Index Using LSTM Models. *Applied Soft Computing*, Elsevier.
- [4] Zhang, Jian, Wang, Xiaoyu, Li, Yifan, and Chen, Wei. (2024). A Hybrid Wavelet–ARIMA–LSTM Approach for Stock Market Forecasting. *Applied Soft Computing*.
- [5] Bollen, Johan, Mao, Huina, & Zeng, Xiao-Jun. (2011). Twitter Mood Predicts the Stock Market. *Journal of Computational Science*, 2(1), 1–8.
- [6] Zou, Jun, Han, Yifan, Li, Xiaofei, Wang, Zhiyong, and Liu, Hong. (2022). Stock Market Prediction via Deep Learning Techniques: A Comprehensive Survey. *IEEE Access*.
- [7] Nabipour, Mohsen, Nayyeri, Payam, Jabani, Hamed, Shahab, Shahaboddin, and Mosavi, Amir. (2020). Deep Learning Applications for Stock Market Prediction. *IEEE Access*, 8, 159438–159447.



**Thank
You**

Q&A